Sine rule

1

a False **b** False **c** True **d** True

2 No, because the sine rule requires one angle to find another angle.

3 Yes, because we can find the opposite angle to c using the angle sum of a triangle. Then we can find the other two sides using sine rule.

4 Yes, because we can find the third angle using the angle sum of a triangle. Then we can find the other two sides using sine rule.

5 No, because we do not know the side opposite $\angle C$, so we do not have an angle-side pair to use to find any other angles.

6

a $\sin A = \frac{x}{b}$ **b** $x = a \sin B$ **c** $\frac{\sin A}{a} = \frac{\sin B}{b}$

Unknown sides

7 $h = 86.52$

8

a $a = 10.5$ **b** $a = 6.9$ **c** $x = 13.1$ **d** $x = 12.9$

e $y = 8.7$ **f** $y = 4.4$

9

a $a = 16.66$ **b** $a = 16.66$

10

a 7.2 mm **b** 19.4 mm

11

a $HK = 733.17 \text{ km}$ **b** $KJ = 444.55 \text{ km}$

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Unknown angles

12

a $x = 30.4^\circ$ **b** $x = 32.7^\circ$ **c** $x = 26.0^\circ$ **d** $x = 43.2^\circ$

13 $14^\circ 47'$

14

a $A = 75.78^\circ$ **b** $b = 3.152\text{m}$ **c** $c = 5.723\text{m}$

15

a $\angle OBA = 39^\circ$ **b** $k = 21.76$

16

a $x = 140.42^\circ$ **b** $x = 133.13^\circ$ **c** $x = 149.12^\circ$

17 $\theta = 140^\circ 40'$

Ambiguous case

18

a $8 < \text{Length} < 10$ **b** $\text{Length} = 8, \text{Length} \geq 10$

c $\text{Length} < 8$

19

a $\theta = 26^\circ$ **b** 8.7385 **c** 9.1247

d No, according to sine rule, $\frac{8.4}{\sin 106^\circ} = \frac{4.0}{\sin 26^\circ}$. Since they are not equal, this triangle is not possible.

20

a No **b** No **c** Yes **d** No

21

a 0 **b** 1 **c** 1 **d** 2

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22

a No **b** No **c** Yes **d** No

e Yes **f** Yes **g** No **h** Yes

i Yes

23

a $x = 41$ **b** 139

24

a

i It does exist **ii** It could be either acute or obtuse.

b

i Does not exist. **ii** No solution

25

a $x = 72.53^\circ$ **b** $x = 107.47^\circ$

26 $3.5 < BC < 5$

Applications

27

a $\angle WST = 136^\circ$ **b** $SW = 234 \text{ km}$ **c** $h = 102.58 \text{ km}$

28

a $\angle ABC = 21.89^\circ$ **b** $\angle BCA = 123.11^\circ$

c $AB = 29.2 \text{ km}$

29

a $\angle A = 38^\circ$ **b** $c = 22.71 \text{ m}$ **c** $b = 22.71 \text{ m}$ **d** 1.9 min

30

a $a = 74^\circ$ **b** $d = 28.35 \text{ m}$ **c** $h = 16.66 \text{ m}$

31

a $\angle ADB = 23^\circ$ **b** $a = 35.65 \text{ m}$ **c** $h = 20.4 \text{ m}$

32

a $\angle ACB = 102^\circ$ **b** $b = 226.55 \text{ m}$ **c** $h = 117 \text{ m}$

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33

a $BC = 307.3\ \text{m}$ **b** $AC = 335.5\ \text{m}$ **c** $41\ \text{m/min}$

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